

Interim Design Report

STM32F0 DISCOVERY HAT



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Executive Summary

This report documents the design process and current design of an air quality monitor STM32F0 Discovery board hardware on top (HAT) intended to detect domestic and industrial gas leakages such as propane, LPG, smoke and hydrogen. The HAT is also integrated with a temperature and humidity sensor to monitor environmental air conditions. All final pcb design files are available on [gitlab repository](#). Appropriate user requirements and an agile design process were applied to develop specifications as outlined in [section 3](#) of this report. Given these use cases this device is capable of detecting combustible gas in the range 300-10000ppm as well as measuring humidity and temperature with a nominal accuracy of $\pm 2\%$ RH and ± 0.2 °C.

The HAT use case is in the aerospace industry where it monitors in-flight conditions. It can also be used in indoor spaces such as science labs in universities, homes, and office buildings to detect any gas leakages and improve individuals' cognitive abilities and reduce respiratory diseases.

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Introduction

This report presents a design of air-quality monitor HAT that integrates with STM32F0 DISCOVERY, simply referred here as STM32DIS. The STM32DIS hat is integrated with a temperature-humidity sensor (SHT30-DIS-B2.5KS) and gas sensor (MQ2) capable of detecting combustible gases such as propane, hydrogen, and LPG with 3ppm resolution. The STM32DIS HAT has a wide range of uses including medical facilities and industrial areas.

A typical application of the HAT is in the aerospace industry where it monitors in-flight conditions. It can also be integrated with indoor spaces such as science labs in universities, homes, and office buildings to detect any gas leakages and improve individuals' cognitive abilities and reduce respiratory diseases.

The HAT gas sensor is designed to withstand temperatures below $20^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and humidity below $65\% \pm 5\%\text{RH}$. Due to the performance drop of the sensor in extreme conditions mentioned above, the design includes a temperature/humidity sensor to monitor the environmental conditions and compensate accordingly.

The HAT board detects combustible gas in the range 300-10000ppm with high sensitivity of R_s (in air)/ R_s (1000ppm isobutane) ≥ 5 and senses temperature and humidity with a resolution of 0.2°C and $2\%\text{RH}$, respectively. The HAT operates off 5V which charges the battery. Sensor and microcontroller interfaces submodules operate off a 3V3 line regulated from the power submodule of the HAT. The HAT has a response time of less than 8 seconds. The HAT interfaces with the STM32DIS. The HAT cost R275.

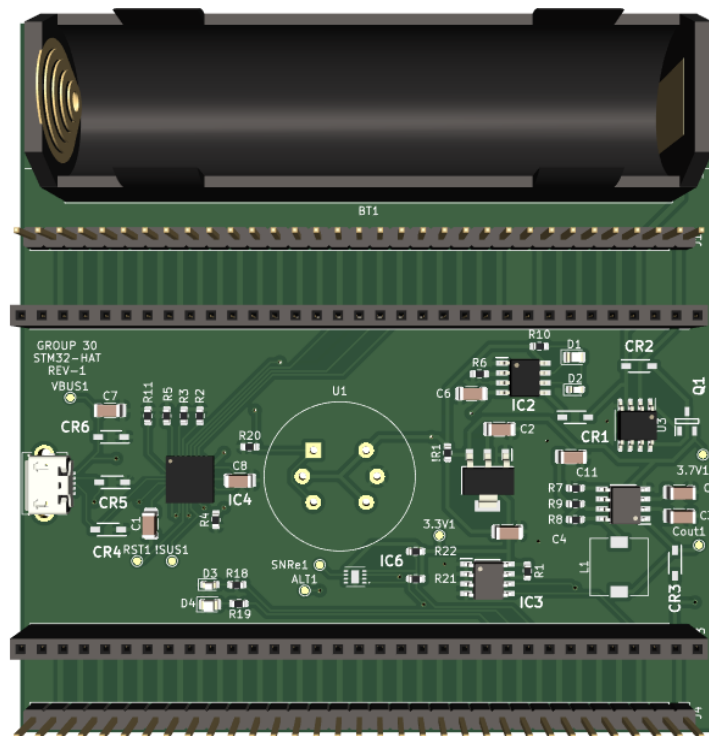


Figure 1: 3D view of HAT

2.1 Project Subsystems Block Diagram

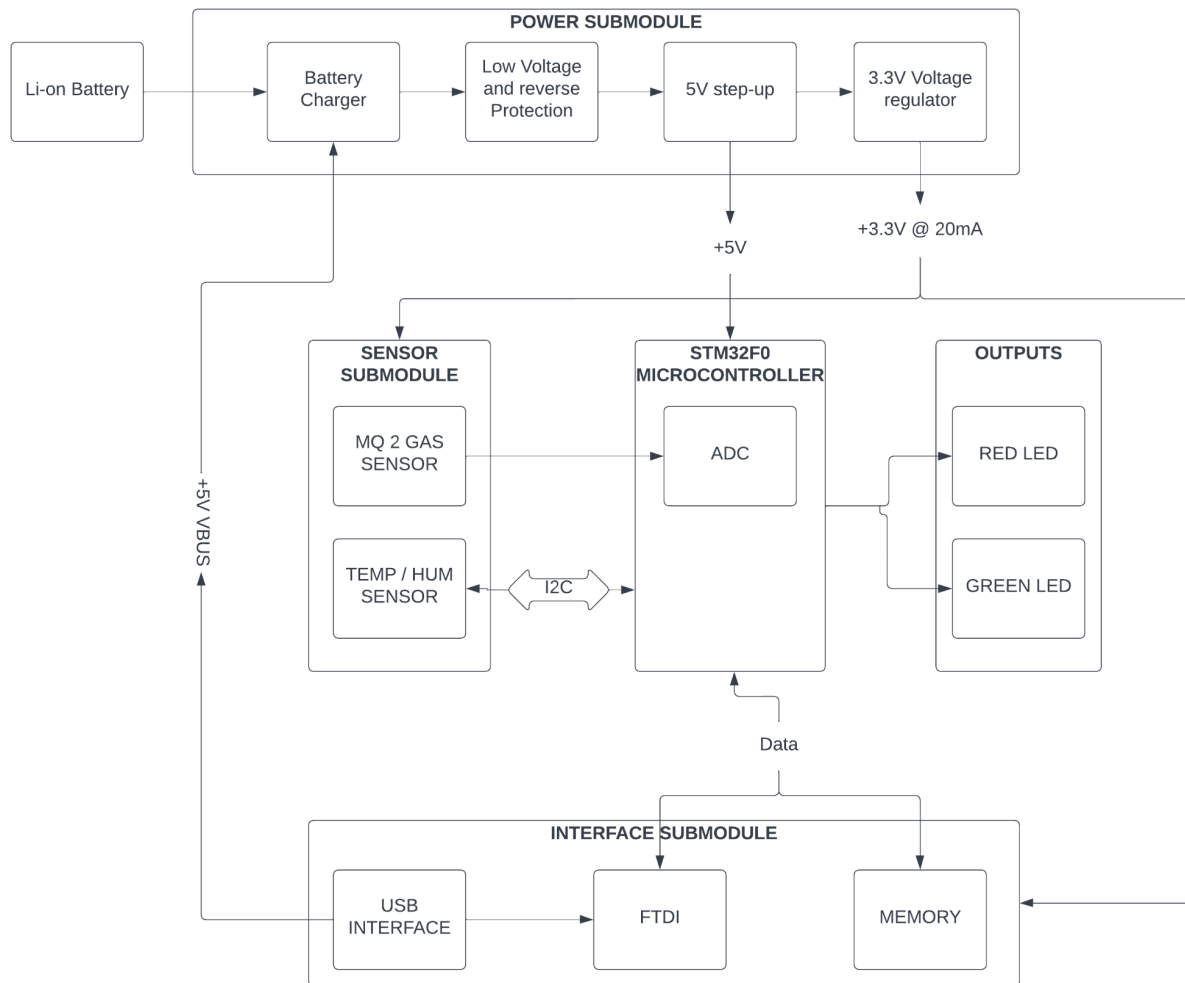


Figure 2: System block

Specifications

3.1 Power subsystem submodule

LION BATTERY CHARGER

- Input voltage -0.3V to 8V
- 4.2V charging voltage with 1.5% accuracy
- 1200mA maximum charging current
- Operating temperature -40 to 85 degrees C
- LED charge status indicator
- TP4056 - ESOP-8_EP_3.9x4.9x1.27P

VOLTAGE STEP-UP

- Input voltage +3.01V to 4.2V
- Line regulation 5V +- 0.012%
- Load regulation +- 0.09%
- Output ripple 500mVpp
- Short circuit current 910mA
- MC34063ADR2G - SOIC-8_3.9x4.9x1.27P

VOLTAGE REGULATOR

- Output voltage 3.3V
- Maximum output current 1A
- Line Regulation: 0.2% Max
- Load Regulation: 0.4% Max
- Operating temperature -40 to 125 degrees C
- AMS1117-3.3 - SOT-223

Reverse polarity protection

- 30V reverse voltage protection
- B5819W - SOD-123

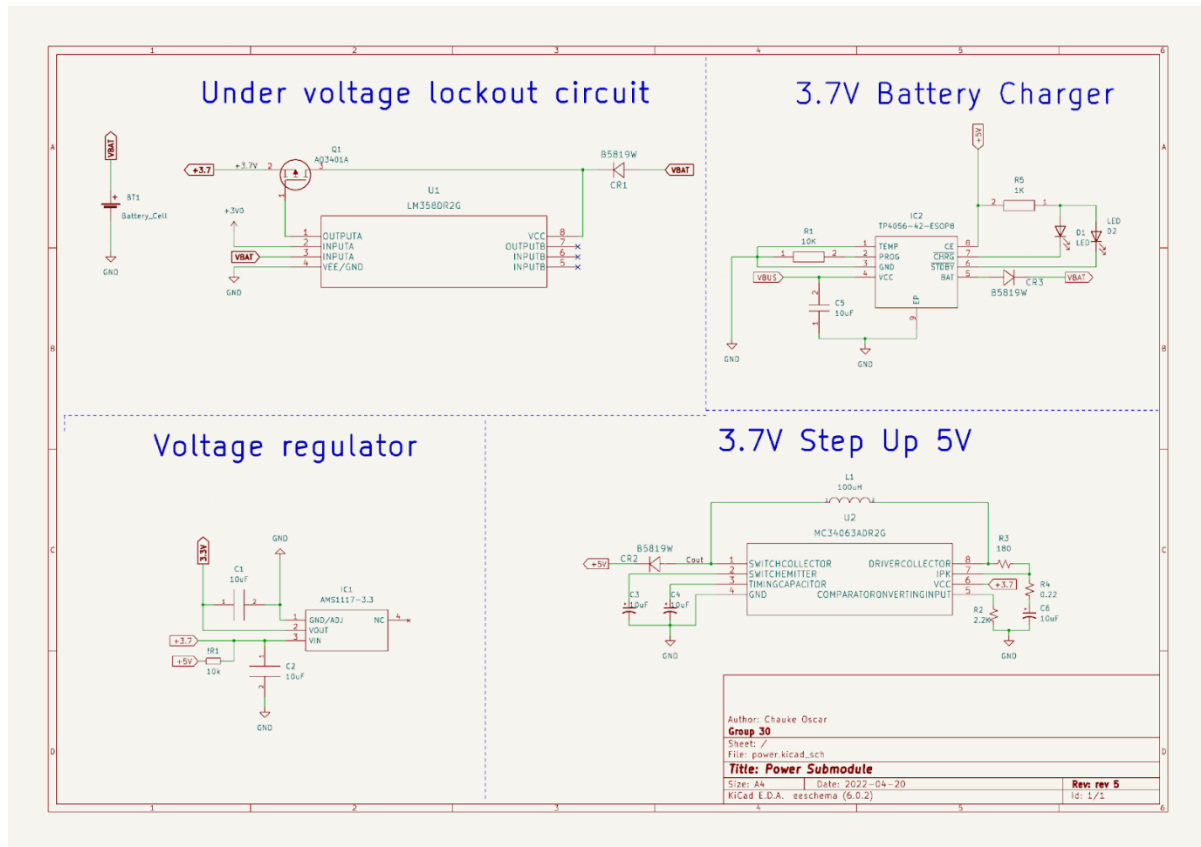


Figure 3: [Power Submodule Schematic \(Gitlab\)](#)

Issues

- The voltage step-up stage has an efficiency of 63%

Comment

All power submodules' requirements are met.

3.2 Microcontroller interfacing submodule

USB INTERFACE

- Contact resistance: 30 mΩ Max.
- Temperature range: -30°C to 80°C
- Total mating force: 3.57 Kgf Max.
- Total unmating force: 1.0 Kgf Max.
- Insulation resistance 100 MΩ Min.
- Dielectric withstanding voltage: 100 V(ac) for 1 minute.
- Current rating: 1,5 PIN 1.8 A Max. 2,3,4 PIN 1 A Max.

MEMORY

- Integrated circuit: AT24C256C-SSHL-T
- Vcc, power supply: +1.7V to +5.5 V.
- Nominal supply current: 2 mA
- Maximum current: 3 mA.
- Maximum standby current: 6 μ A.
- Input leakage current: 0.1 μ A
- Maximum input leakage: 3.0 μ A.
- Output leakage current: 0.05 μ A
- Maximum output leakage current: 3.0 μ A.

FTDI

- Integrated circuit: CP1202
- Voltage supply: +3.0 V to +3.6 V
- Temperature of operation: -40°C to +80°C.
- Thermal resistance: 32 °C/W.
- Normal supply current: 20mA
- Maximum current: 26mA.

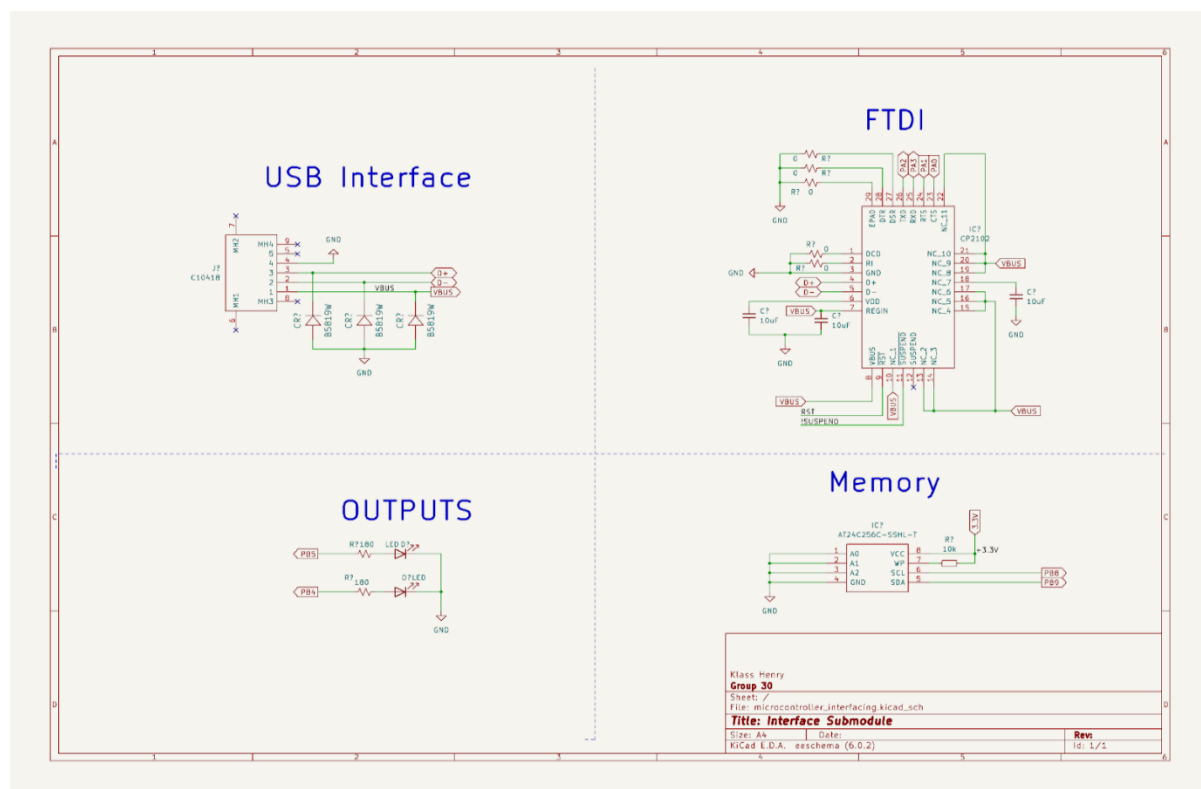


Figure 4: [Microcontroller Interface submodule schematic \(Gitlab\)](#)

Comment:

- 1) All microcontroller interfacing submodule requirements are met.

3.3 Sensing submodules

HUMIDITY AND TEMPERATURE SENSOR

- IC used: SHT30-DIS-B2.5KS
- Supply voltage range (VDD), from 2.15V to 5.5V
- Max supply current of 1.5mA when the sensor is measuring data
- Interface with communication speeds up to 1 MHz
- Typical accuracy of 2%RH and 0.2 °C
- Response time of 8s
- Operating temperature range from -40 to 125 °C
- Package: DFN-8_EP_2.5x2.5x0.5P
- Mounting type: Surface mount
- Output type: Digital
- Heater power ranges from 3.6mW to 33mW
- Input current rating on any pin ± 100 mA

SEMICONDUCTOR SENSOR FOR COMBUSTIBLE GAS

- Model No. MQ-2
- Sensor type: Semiconductor
- Operating voltage: 5V0.1 V
- Concentration scope: 300-10000ppm (combustible gas)
- Output type: Analog
- Load resistance: Adjustable
- Heater resistance: $31\Omega \pm 3\Omega$ (Room temperature)
- Heater consumption: ≤ 900 mW
- Sensing resistance: $2K\Omega$ - $20K\Omega$ (in 2000ppm isobutane)
- Preheat time: Over 48 hours

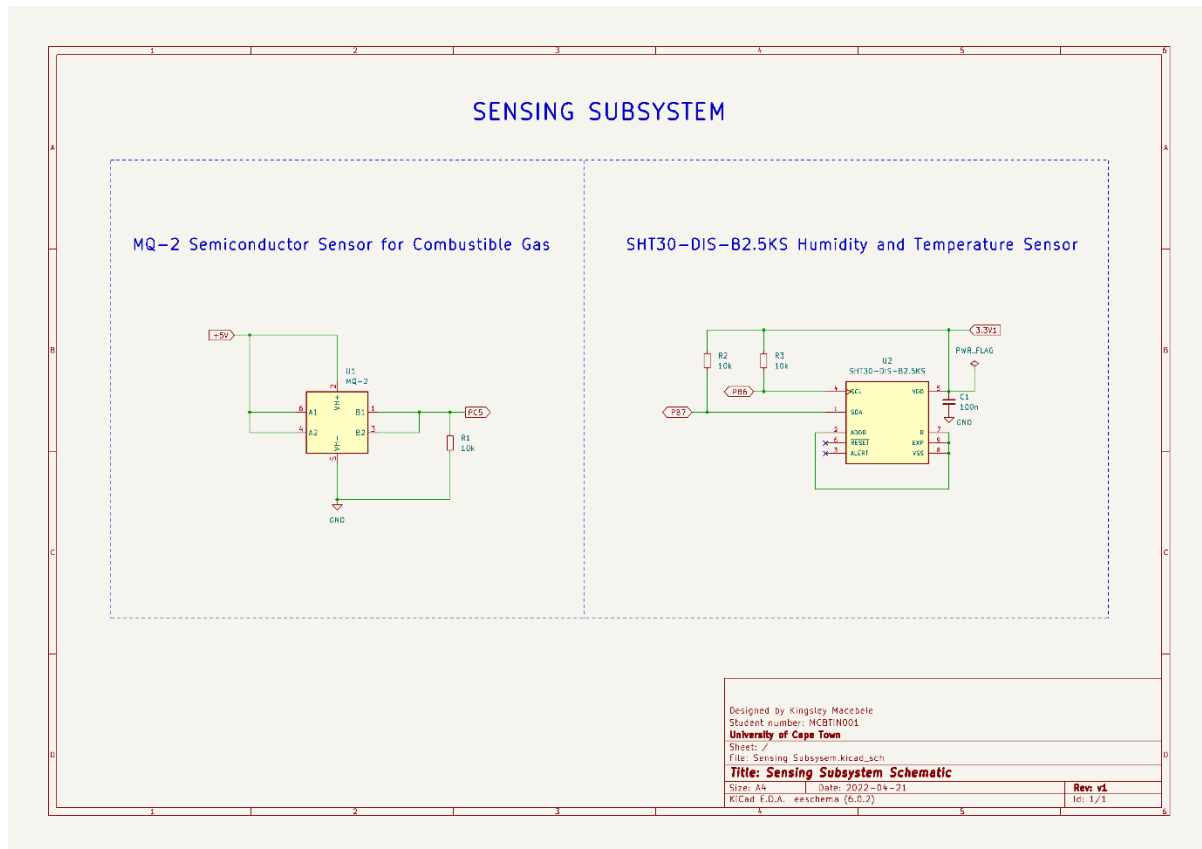


Figure 5: [Sensor submodule schematic\(Gitlab\)](#)

Issues

- The MQ2 does not operate off the 3V3 line as stipulated in the design requirements, it operates on a 5V supply voltage.

Comments

The sensing submodule is a good design because an agile design process was implemented to reach the final design of the sensing submodule.

Power Budget Analysis

4.1 Power submodule

Component	Supply Voltage (nominal V)	Minimum Current (mA)	Typical Current (mA)	Maximum Current (mA)	Minimum Power (mW)	Typical Power (mW)	Maximum Power (mW)
LM358DR2G	5	20	40	40	100	200	200
TP4056	4	0.055	0.15	0.5	0.22	0.6	2
MC34063ADR2G	3.7	100	100	100	370	370	370
AMS1117-3.3	4.8	5	5	11	24	24	52.8
Totals		125.055	145.15	151.5	494.22	594.6	624.8

Note: For MC34063ADR2G, the current used in calculations is the maximum current. Minimum current depends on the STM32DIS current requirements at different operations.

4.2 microcontroller interfacing submodule

Component	Supply Voltage (nominal V)	Minimum Current (mA)	Typical Current (mA)	Maximum Current (mA)	Minimum Power (mW)	Typical Power (mW)	Maximum Power (mW)
CP2102	5	20	40	40	100	200	200
AT24C256C	4	0.055	0.15	0.5	0.22	0.6	2
Totals		20.055	40.15	40.5	100.22	200.6	202

Note: Power dissipated by usb interface is negligible therefore not included in power budget.

4.3 Sensing submodules

Component	Supply Voltage (nominal V)	Minimum Current (mA)	Typical Current (mA)	Maximum Current (mA)	Minimum Power (mW)	Typical Power (mW)	Maximum Power (mW)
SHT30-DIS-B 2.5KS	3.3	0.6	0.6	1.5	1.98	1.98	4.95
MQ-2	5	180	180	180	900	900	900
Totals		180.6	180.6	181.5	901.98	901.98	904.95

Note: The current used in calculation is the maximum current, the current drawn by MQ2 depends on the environmental conditions.

4.4 System Total

Submodule	Supply Voltage nominal V)	Minimum Current (mA)	Typical Current (mA)	Maximum Current (mA)	Minimum Power (mW)	Typical Power (mW)	Maximum Power (mW)
Power Submodule	3.7	125.055	145.15	151.5	494.22	594.6	624.8
Sensing Submodule	5	180.6	180.6	181.5	901.98	901.98	904.95
Microcontroller Interface Submodule	5	20	42	43	100	206.6	209.9
Totals		325.655	367.75	376	1628.275	1838.75	1880

Design process

An agile design process was followed to reach the current final design. Analog MQ-2 Semiconductor Combustible Gas sensor was chosen due to a shortage of digital SGP30-2.5K TVOC and CO2 Sensor on JLCPCB. As a result, an analog temperature/humidity sensor (SHT30-ARP-B2.5KS) was changed to a digital temp/hum sensor (SHT30-DIS-B2.5KS) to meet the design requirements. Integrated circuits were used for every stage to minimize error and increase chances of the board to function. MC34063ADR2G was chosen for the voltage step-up stage with the substandard efficiency of 63% when operating at 5V as it was the only basic part on the JLCPCB part list. Removed redundant features like voltage regulator and an analog temp/hum sensor to minimize the costs.

Conclusion

Agile (iterative) design process was followed whereby each component of the submodules were chosen to meet the design requirements specified by the client. This resulted in removing redundant features of each submodule to minimize the costs. The STM32F0 Discovery HAT accurately detects gas leakages in indoor areas thus achieving the envirosensing requirements of the user. The HAT contains three submodules, the microcontroller interface, power and sensing. For efficiency, each of the submodules uses integrated circuits. To ensure the HAT operates without a high chance of fault, it has been designed in such a way that it operates within the specifications of the different components used. There's a limit to the costs used to ensure affordability of the HAT amongst the poor. The total power used by the HAT is enough to ensure it is operating well. Data from the sensors has digital output in the form of two LEDs, one that is green for good quality, and a red LED to indicate air quality that is inferior. A permissive license (MIT open source license) has been used to allow users to modify and use the HAT without restrictions.